

Course Code: CSE2013	Course Title: Information Theory and Coding	TPC	4	0	4
Version No.	2.0				
Course Pre-requisites/ Co-requisites	MAT1011				
Anti-requisites (if any).	None				
Objectives:	1. To define and apply the basic concepts of information theory to the communication systems 2. To study different types of communication channels 3. To define the fundamental limit of compression without information loss. 4. To study various data compression methods and describe the most common such methods 5. To understand the theoretical framework upon which error-control codes are built				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement	PO's / PEO's			
CO1	Quantify the notion of entropy, joint entropy, relative entropy, conditional entropy, and channel capacity of a system in a mathematically sound way	PO1, PO2, PEO1			
CO2	Explain the significance of mutual information in the communication systems.	PO2, PO3, PEO1, PEO2			
CO3	Implement the various types of source coding algorithms and analyze their performance	PO2, PO4, PEO1			
CO4	Able to understand the encoding and decoding schemes of linear and cyclic codes and the significance of minimum Hamming distance	PO4, PO5, PEO2,			
CO5	Explain the basic difference between block codes and convolutional codes and Viterbi decoding algorithm.	PO7, PO11, PEO2			
CO6	Analyse the concepts mathematically with examples	PO11, PO12, PEO3			
		TOTAL HOURS OF INSTRUCTIONS: 60			
Module No. 1	Introduction			6 Hours	
Introduction to Information Theory, Information rate and entropy, Measure of Information, Properties of entropy of a binary memory less source.					
Module No. 2	Conditional and Joint Entropy, and Mutual Information			10 Hours	
Joint entropy, Conditional entropy, Mutual information, Discrete memoryless channels - BSC, BEC, noise-free channel, Channel with independent I/O, Cascaded channels. Channel Capacity.					
Kullback–Leibler divergence (Relative Entropy), Cross-Entropy, Jensen–Shannon divergence					
Module No. 3	Lossless Compression Techniques			12 Hours	
Source Coding Theorem, Uniquely Decodable codes, Instantaneous Codes, Kraft-McMillan Inequality, Shannon Fano coding, Huffman coding, Minimum variance Huffman coding, Arithmetic coding.					
Dictionary Coding- Lempel-Ziv-Welch (LZW) coding.					

Module No. 4	Linear Codes	12 Hours
Channel coding theorem for DMC, Code Rate, Hamming weight, Hamming distance, Minimum Hamming Distance.		
Block Codes – Introduction to Linear Block Codes, Generator Matrix, Parity Check Matrix, Minimum Hamming Distance from Parity check matrix. Error-detecting and Error-correcting capability, Syndrome Decoding, Standard Array, Probability of an Undetected Error.		
Hamming Codes, Single Parity Check Codes, Dual Codes.		
Module No. 5	Cyclic Codes	8 Hours
Cyclic codes – Generator polynomial, Parity check polynomial, Generator and Parity check matrices, Encoding of cyclic codes, Syndrome calculation, Encoder and decoder – CRC		
Module No. 6	Convolutional Codes	12 Hours
Convolutional codes – code tree, trellis, state diagram-encoding-decoding. Maximum Likelihood Detection, Sequential search and Viterbi algorithm – Principle of Turbo coding, Interleaved convolutional codes,		
Text Books		
1. J. A. Thomas, “Elements of Information Theory”, T. M. Cover, 2e, Wiley, 2013.		
2. R. Togneri, C.J.S deSilva, “Fundamentals of Information Theory and Coding Design”, 1e, CRS Press, Imprint: Taylor and Francis, 2003.		
3. Shu, Lin. Error control coding. Pearson Education India, 2011.		
References		
1. Haykin, Simon. Communication systems. John Wiley & Sons, 2008.		
2. R. Bose, “Information Theory Coding and Cryptography”, 3e paperback, Tata McGraw Hill, 2016.		
Mode of Evaluation	CAT-1	20%
	CAT-2	20%
	FAT	20%
	QUIZ/ ASSIGN.	40%
Recommended by the Board of Studies on		
Date of Approval by the Academic Council		